Cardiology of Birds
VET437
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Birds evolved from theropod dinosaurs of the Late Jurassic. So, technically dinosaurs are not extinct, and this lecture could be renamed 'Cardiology of Dinosaurs' ☺. This evolutionary link, and new techniques in paleontology that help scientists glean metabolic data from the fossil record also support that dinosaurs were warm blooded!

Anatomy of the Avian Heart

Birds are endothermic and have high aerobic demands due to flight, which requires very efficient oxygen transport. This is achieved in part by a larger (relative to body mass) heart, as compared to mammals. The heart rate relative to body weight is actually lower in birds as compared to mammals, but their relative heart size and a large capacity to increase heart rate during flight (2 to 4 times) overcomes this discrepancy. Because of a high stroke volume (SV), the cardiac output (CO) of birds is high (CO = SV x HR). Recall that mean arterial pressure (MAP) is related to CO x systemic vascular resistance (SVR). The higher CO is counterbalanced to some degree by a lower SVR, yet overall, the MAP of birds is higher when compared to mammals.

Like mammals, birds have a four chambered heart that connects the two circulations (systemic and pulmonary) in series. The right ventricle wraps around the conical left ventricle. Relative ventricular wall thicknesses reflect the vascular resistance (can be approximated by blood pressure) in the respective circulation, with left ventricular systolic pressures being 4 to 5 times that of pulmonary systolic pressure and the left ventricle being 2 to 3 times thicker than the right ventricle. The relative heart size of birds varies due to species differences and fitness levels (flighted versus non and captive versus free ranging).

Anatomical differences from dogs and cats are briefly discussed here and further emphasized with italics. Birds have *paired cranial vena cavae* and a single caudal vena cava bring venous blood back to the right atrium. The right atrioventricular (AV) valve (tricuspid valve) is *muscular* and connected to the roof of the right ventricle via a muscle bundle and to the interventricular septum by a narrow membrane. There is variation amongst species where and how the pulmonary veins coalesce and drain into the left atrium. The avian aorta arises from the *right 4th aortic* arch as compared to the 4th left aortic arch in mammals. If the aortic arch is seen on radiographs it will curve toward the right, as compared to the left in dogs and cats. The *aortic valve has a sphincter-like ring of myocardium* that likely plays a role in valve function. Immediately *two large brachiocephalic trunks* (can be larger than the ascending aorta itself in flighted birds) branch off to supply arterial blood to cranial structures and the wings. Differences in the avian conduction system include the anatomy of the bundle branches and the depth of Purkinje fiber penetration. Birds have a left and right branch and a *figure 8 shaped middle branch* that wraps around the aorta and the right AV valve. Birds have a more *deeply penetrating Purkinje system (type B)*, which leads to a more rapid (burst) depolarization.

Prevalence of Cardiovascular Disease

Poultry and companion psittacine birds have the highest documented prevalence of cardiovascular disease. Modern commercial poultry have been genetically selected for fast growth rates and are managed intensively to achieve high feed conversion ratios. This increases the workload on the cardiovascular system and can predispose these birds to right ventricular heart failure, arrhythmias, and sudden death. The pathophysiology may include a relatively inadequate left ventricle and a bi-ventricular failure where right-sided failure is driven by post-capillary pulmonary hypertension (secondary to increased pulmonary venous pressure) and pre-capillary pulmonary hypertension due to chronic hypercapnia. Primary and secondary (to nutritional deficiency, drug related) dilated cardiomyopathy has also been reported in production birds.

In pet birds (especially psittacines), atherosclerosis is the underlying cause of most noninfectious cardiovascular disease. Other cardiovascular diseases include dilated cardiomyopathy (primary/genetic versus secondary to ischemia [atherosclerosis], toxin, nutritional deficit, or infectious agent), hypertrophic cardiomyopathy (primary seems uncommon, probably more often secondary to atherosclerosis and aortic valve stenosis or systemic hypertension), degenerative valve disease (usually mitral and/or aortic), pericardial disease, and cardiac masses (neoplasia or granuloma). Congenital heart disease is not as common as acquired heart disease. Congenital diseases that have been described include aortic valve stenosis/hypoplasia, ventricular and atrial septal defects, patent ductus, arteriosus, pulmonary valve stenosis, and truncus arteriosus (the aorta and pulmonary artery share a common trunk).

History and Cardiovascular Exam

It is important to determine the signalment (noting the species, gender, and age) and document a thorough history with a careful survey of the diet, appetite, physical activity, medications, supplements, and any exposures to toxins. Open ended questions about the patient's general demeanor, activity level, breathing, and abdominal distention may identify signs consistent with cardiovascular disease. Cardiovascular disease should be included on the differential diagnosis list when nonspecific signs such as lethargy, apathy, and inactivity are reported. Similarly respiratory abnormalities have many differential diagnoses, and heart disease should be one of them. Delayed absorption (>48 hours) of subcutaneous fluid is supportive of circulatory overload and overt or impending heart failure.

Evaluate the bird's demeanor, posture, mobility, general body condition, and breathing prior to handling. It is not uncommon for cardiovascular disease to go undiagnosed for many years and patients may present with very advanced disease and severe clinical signs. If the bird is in respiratory distress, oxygen supplementation is indicated to maintain stability while you complete your exam. If the respiratory distress is severe, place the bird in an oxygen cage and only proceed with your exam as their stability allows. Keep the bird upright during your exam. Mucous membranes should be moist and pink; however, pigmentation may make their assessment difficult. Sometimes cyanosis can be appreciated at the periorbital skin. Peripheral arterial pulse palpation can be performed at the superficial ulnar artery, which is located at the medial aspect of the proximal antebrachium. Evaluate the periorbital area and feet for peripheral edema (uncommon) and look at the bird's overall shape to assess for coelomic distension. Also feel the extremities to subjectively assess their warmth. Increased lung sounds may be heard when pulmonary edema is present; however, crackles are not typically heard.

Cardiac auscultation in birds should start at the base of the keel on the left and right sides. Move your stethoscope along each side of the keel and to the thoracic inlet to identify where heart sounds are heard best. As the heart rate is normally elevated and further elevated by the stress of handling, auscultation is difficult and often insensitive. Listen for an irregular rhythm and breaks/pauses in the rhythm, as this can occur due to premature beats or inappropriate pauses. Soft murmurs (< grade III/VI) may be difficult to hear at high heart rates, yet moderate and loud murmurs can often be heard. Though timing usually cannot be fully elucidated, more common murmurs include mitral regurgitation (systolic) and aortic insufficiency (diastolic).

There are multiple differential diagnoses for coelomic distension. One of these is hepatomegaly and/or ascites due to systemic venous congestion secondary to heart failure. Severe ascites can compress the air sacs and contribute to respiratory distress – so if the patient is having respiratory difficulty in the face of a taught, distended coelom, centesis may be necessary for stabilization prior to moving on with your exam or other diagnostic tests.

Patients with suspected cardiovascular disease should have a baseline diagnostic workup, including serum biochemistry, CBC, radiographs, point of care ultrasound, echocardiography, blood pressure measurement, and electrocardiography). Advanced imaging (CT, MRI, endoscopy) is sometimes employed to diagnose and characterize cardiovascular diseases in birds. Cytology of coelomic effusions secondary to congestive heart failure is usually consistent with a pure or modified transudate.

Radiography
Orthogonal (usually ventrodorsal [VD] and lateral projections) radiographs allow for the evaluation of the cardiac silhouette size and shape. Some normal 'tips' have been derived for specific species. For example, in the Amazon parrot, the cardiac silhouette measured across the base is about 50% of the width of the coelomic cavity on the VD view. Please see the PPT slides for some additional metrics. An enlarged cardiac silhouette and/or increased pulmonary density are indications for echocardiography for a more detailed evaluation of the cardiac structure and function. Differentials for an enlarged cardiac silhouette include cardiomyopathy (primary or secondary), volume overload (valvular insufficiencies from any cause), masses, and pericardial effusion.

The great vessels, including the aorta, brachiocephalic trunks, and pulmonary arteries and veins are often visible. In larger birds, the common carotid and axillary arteries may be seen on the VD view. Increased radiodensity and tortuosity suggest (but do not confirm) atherosclerosis. The more definitive findings of focal or linear mineralization (most often found in the ascending and descending aorta and brachiocephalic trunks) is highly supportive of advanced atherosclerosis. Increased density of the reticular pattern may be seen with pulmonary edema yet can also be due to other interstitial diseases (inflammation, fibrosis, or neoplastic). The hepatic silhouette may be enlarged secondary to venous congestion or due to summation of an enlarged cardiac apex (cardiomegaly or pericardial effusion). Enlargement of the cardiac and hepatic silhouettes may lead to caudal displacement of adjacent GI tract or air sac compression. Ascites will cause loss of coelomic detail.

Electrocardiography
An abnormal rate or rhythm diagnosed during physical exam, further investigation of collapse events, and anesthetic monitoring are often indications to perform an electrocardiogram (EKG). Whether or not to sedate – or even anesthetize – a patient will depend on the patient's demeanor and tolerance for handling. The EKG in birds should be recorded using the fastest paper speed available (≥ 50mm/sec). The gain can also be adjusted as necessary, typically starting at 10mm/mV. Flat clips or needle electrodes are applied to the left and right propatagium and left and right legs (inguinal skin web near the stifle; RF – white; LF – black; RR – green; LR – red). Birds have a type B (deeply penetrating) Purkinje system. The ventricles are depolarized very rapidly and have a mean vector of depolarization that is usually around -90° (aligns with the negative electrode of aVF). Therefore the 'QRS' of birds is predominantly a deep negative S wave in lead II and isoelectric in lead I (which is perpendicular to aVF). In birds, the Q is usually absent, and a small r wave is present before the large S wave (rS morphology). P and T wave morphologies are similar to other small animal species. Due to the fast heart rate of birds, the P may be in the preceding T wave (P on T). We will review several arrhythmias and their treatment in the lecture.

Ultrasonography
Point of care ultrasound (POCUS) is a valuable tool to quickly diagnose cavitary (coelomic or pericardial) effusions. Usually POCUS can be performed 'cage-side' in awake or sedated patients. Fluid will appear black (anechoic) within these spaces. With training, abnormalities in cardiac structure and function and hepatic parenchyma changes may be identified with POCUS. The echocardiographic window for the heart is usually immediately ventral to the keel (ventromedian approach). Parasternal windows are possible in some species. The two major echocardiographic views are the horizontal view (roughly similar to the left apical 5-chamber view) and the vertical view (a two-chamber view prioritizing the LA/LV). Two-dimensional imaging allows for assessment of chamber dimensions and function and valve morphology and mobility. Color and spectral Doppler allow for evaluation of valve competency and function, and pressure gradients. Further information about the full echocardiographic exam and additional advanced imaging techniques can be found in Chapter 6 of Current Therapy in Avian Medicine and Surgery: Brenna Colleen Fitzgerald & Hugues Beaufrère.

Blood pressure measurement
Direct blood pressure measurement is the most accurate approach yet requires catheterization of an artery. Instead, an indirect (Doppler) method of estimating blood pressure is used in the clinic. Doppler blood pressure measurement is performed by placing the ultrasound crystal over the superficial ulnar artery or cranial tibial artery. A cuff whose width is 30 to 40% of the limb's circumference is placed proximal to the point of measurement and connected to a sphygmanometer, which is inflated until pulsatile flow ceases. The cuff is deflated and the pressure at which the first pulsatile sound returns in the systolic blood pressure estimate. This indirect method of assessing blood pressure is poorly correlated to direct pressure measurement. However, measurements consistently obtained during a procedure may be useful to monitor trends.

In lecture, we will briefly discuss several causes of cardiovascular disease in birds (with a focus on atherosclerosis). We'll also look at a few examples of arrhythmias. Finally, we'll discuss the approach to treatment of these diseases.